

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1680

Bayou Vista 100 kV
January 30, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1680. The Interconnection Customer proposes a 100.7 MW battery energy storage facility interconnecting at the Bayou Vista 100 kV substation in the Illinois area. The request was received on February 25, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 619 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,684,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: North Vly-W Danville 15 100kV
Contingency: Weyerhaeuser Co-N Vly 100kV Line
Loading: 153.0% (Rating: 129 MVA)
DFAX: 9.0%
Pre-project loading: 146.0%
2. Facility: Lawrence Hill-North Valley 100kV
Contingency: East Jackson 1&2-Dyckesville 100kV
Loading: 135.5% (Rating: 209 MVA)
DFAX: 10.0%
Pre-project loading: 130.7%
3. Facility: E Jackson 1&2-Allaattah 100kV Cir. 1
Contingency: E Jackson 1&2-Allapattah 100kV Cir. 2
Loading: 122.3% (Rating: 142 MVA)
DFAX: 22.3%
Pre-project loading: 106.5%
4. Facility: E Jackson 1&2-Allapattah 100kV #2
Contingency: E Jackson 1&2-Allapattah 100kV
Loading: 122.3% (Rating: 142 MVA)
DFAX: 22.3%
Pre-project loading: 106.5%
5. Facility: Wildwood Substation-Blue Island 100kV Cir. 2
Contingency: Wildwood-Blue Island 100kV
Loading: 120.5% (Rating: 165 MVA)
DFAX: 25.1%
Pre-project loading: 105.2%

6. Facility: Wildwood-Blue Island 100kV
Contingency: Wildwood-Blue Island Substation 100kV Cir. 2
Loading: 115.1% (Rating: 173 MVA)
DFAX: 25.1%
Pre-project loading: 100.4%
7. Facility: E Jackson 1&2-Dyckesville 100kV Line
Contingency: Shoreham Power Station-Castle Afb 100kV
Loading: 114.1% (Rating: 204 MVA)
DFAX: 13.6%
Pre-project loading: 107.4%
8. Facility: Blue Island-Bayou Vista 100kV
Contingency: Bayou Vista-Blue Island Substation 100kV Circuit 1
Loading: 111.8% (Rating: 148 MVA)
DFAX: 49.3%
Pre-project loading: 78.2%
9. Facility: Franz Holmes Substation-Sand Ridge 100kV
Contingency: Lansing-Dyckesville 100kV L
Loading: 110.6% (Rating: 138 MVA)
DFAX: 8.1%
Pre-project loading: 104.7%
10. Facility: Robbins Community Power Llc-Sand Bar 100kV Cir. 2
Contingency: Robbins Community Power Llc-Sand Bar 100kV Cir. 1
Loading: 108.4% (Rating: 161 MVA)
DFAX: 13.1%
Pre-project loading: 100.2%
11. Facility: Weyerhaeuser Co-North Delphos 100kV line
Contingency: West Danville 15-North Valley 100kV
Loading: 107.6% (Rating: 131 MVA)
DFAX: 16.9%
Pre-project loading: 94.7%
12. Facility: Lansing-Dyckesville 100kV Line
Contingency: Sand Ridge-Castle Afb 100kV Line
Loading: 106.4% (Rating: 302 MVA)
DFAX: 13.5%
Pre-project loading: 101.9%
13. Facility: Sand Ridge-Castle Afb Substation 100kV L
Contingency: Lansing-Dyckesville 100kV
Loading: 104.5% (Rating: 270 MVA)
DFAX: 10.6%
Pre-project loading: 100.6%
14. Facility: Bayou Vista-Blue Island 100kV Ckt 1
Contingency: Bayou Vista-Blue Island 100kV
Loading: 103.9% (Rating: 159 MVA)
DFAX: 49.3%
Pre-project loading: 72.7%
15. Facility: Shoreham Power Sta-Castle Afb 100kV
Contingency: E Jackson 1&2-Dyckesville Substation 100kV
Loading: 102.9% (Rating: 212 MVA)
DFAX: 7.3%
Pre-project loading: 99.5%
16. Facility: 9202-9206-Robbins Community Power Llc 100kV
Contingency: West Danville 15 Substation-Blakfoot Substation 230kV

Loading: 102.4% (Rating: 133 MVA)

DFAX: 15.0%

Pre-project loading: 91.0%

17. Facility: Sand Bar-Wildwood 100kV Ckt 1

Contingency: Wildwood-Sad Bar 100kV Cir. 2

Loading: 101.6% (Rating: 174 MVA)

DFAX: 13.6%

Pre-project loading: 93.8%

18. Facility: Wildwood Substation-Sand Bar 100kV

Contingency: Wildwood-Sand Bar 100kV Line 1

Loading: 101.6% (Rating: 174 MVA)

DFAX: 13.6%

Pre-project loading: 93.8%

19. Facility: Blue Island-Allapattah Substation 100kV

Contingency: Blue Island-Allapattah 100kV Line

Loading: 101.4% (Rating: 140 MVA)

DFAX: 20.4%

Pre-project loading: 86.7%

20. Facility: Allapattah-Blue Island 100kV Cir. 2

Contingency: Blue Island-Allapattah 100kV Line 1

Loading: 101.4% (Rating: 140 MVA)

DFAX: 20.4%

Pre-project loading: 86.7%

21. Facility: Robbins Community Power Llc-Sand Bar 100kV Line 1

Contingency: Robbins Community Power Llc-Sand Bar 100kV

Loading: 100.2% (Rating: 174 MVA)

DFAX: 13.1%

Pre-project loading: 92.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
West Danville 15-N Vly 100kV	Rebuild line with higher-capacity conductor	\$3,725,000	5% (\$200,000)
Lawrence Hill-North Valley Substation 100kV line	Rebuild line with higher-capacity conductor	\$1,450,000	4% (\$63,000)
E Jackson 1&2-Allaattah 100kV Cir. 1	Rebuild line with higher-capacity conductor	\$1,260,000	44% (\$559,000)
E Jackson 1&2-Allapattah Substation 100kV Circuit 2	Rebuild line with higher-capacity conductor	\$1,144,000	44% (\$508,000)
Wildwood Substation-Blue Island 100kV Cir. 2	Rebuild line with higher-capacity conductor	\$1,258,000	50% (\$631,000)
Wildwood-Blue Island 100kV Ckt 1	Rebuild line with higher-capacity conductor	\$1,843,000	50% (\$924,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E Jackson 1&2-Dyckesville 100kV	Rebuild line with higher-capacity conductor	\$1,999,000	9% (\$182,000)
Blue Island-Bayou Vista 100kV	Rebuild line with higher-capacity conductor	\$1,655,000	65% (\$1,071,000)
Franz Holmes Substation-Sand Ridge 100kV	Rebuild line with higher-capacity conductor	\$1,388,000	15% (\$208,000)
Robbins Community Power Llc-Sand Bar 100kV Cir. 2	Reconductor line	\$770,000	34% (\$261,000)
North Delphos-Weyerhaeuser Co 100kV	Reconductor line	\$602,000	27% (\$164,000)
Lansing-Dyckesville 100kV Line	Reconductor line	\$486,000	8% (\$39,000)
Sand Ridge-Castle Afb Substation 100kV L	Reconductor line	\$248,000	7% (\$18,000)
Blue Island-Bayou Vista Substation 100kV	Reconductor line	\$387,000	65% (\$250,000)
Castle Afb-Shoreham Power Station 100kV	Reconductor line	\$973,000	6% (\$57,000)
Robbins Community Power Llc-9202-9206 100kV	Reconductor line	\$851,000	32% (\$271,000)
Sand Bar-Wildwood 100kV Ckt 1	Reconductor line	\$663,000	34% (\$224,000)
Wildwood Substation-Sand Bar 100kV	Reconductor line	\$589,000	34% (\$199,000)
Blue Island-Allapattah Substation 100kV	Reconductor line	\$1,356,000	53% (\$717,000)
Allapattah-Blue Island 100kV Cir. 2	Reconductor line	\$1,623,000	53% (\$858,000)
Robbins Community Power Llc-Sand Bar 100kV Line 1	Reconductor line	\$828,000	34% (\$280,000)
	Total allocated to project		\$7,684,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.